

Novel Circuit Architecture for configurable eDP and MIPI DPHY IO

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Abstract— Display interfaces are key components in present day System on Chips (SOC) as display provides maximum user experience in any product. SOC's used for smart phones, tablets or mobile segments like laptops use Embedded Display Port (eDP) interface or MIPI Display Serial Interface (DSI) for driving display panels. Most SOC's integrate both eDP PHY (Physical layer) and MIPI DPHY to enable SOC reusability across smallest display (MIPI DSI) application to high resolution display (eDP) use cases. As a result, one PHY is unused in most SOC based on application. This paper describes Input output (IO) Architecture & circuit topology to integrate eDP and MIPI DPHY. The paper also describes circuit Architecture to achieve integrate PHY using thingate devices to enable design in sub nm (nanometer) process technology, this paper explains implementation and post silicon results eDP-MIPI Combophy.

Keywords—Embedded display, eDP, MIPI DSI, DPHY, Display port

I. INTRODUCTION

eDP MIPI Combophy is a very salient feature which enables the original Equipment manufacturers (OEM) to choose the embedded panel of their wish. This also enables cost savings in terms of die area and bumps. This could be enabled in the process technology where it supports thick gate devices but restricts the eDP data rates to up to 2.7Gbps. To enable 4K/5K/8K embedded panels, eDP data rate required will be 5.4Gbps/8.1Gbps which cannot be designed with the thick gate. Some latest process technology nodes may not support the thick gate devices due to cost or performance issues. Hence to overcome both the combo performance and lack of thick gate support in the process technology, the Novel eDP 1.5 and MIPI DSI 2.0 Combophy solution with thin gate-based design is proposed.

II. COMBO PHY ARCHITECTURE

eDP display PHY needs 4 Transmitters (TX) and one Auxiliary (AUX) port to communicate display protocol over display port cable. By eDP1.5 specification each lane can operate up to 8.1Gbps to support high resolution display up to 4K 30hz refresh rate. MIPI DPHY is display interface used for low power and handheld devices which also has 4 TX and 1 CLK lane. By MIPI DPHY2.1 specification TX can operate at 4.5 Gbps. Basic idea of this paper is to integrate eDP and MIPI DPHY on single PHY and provide configurability to OEMs to choose the interface by application. Fig1 below shows the high-

level architecture showing the eDP or MIPI DPHY configuration by lanes.

In eDP mode TX0, TX1, TX2, TX3 operate as data lanes and AUX lane works as bi-directional side-band communication. In case of MIPI DPHY, AUX lane is configured as data lane 0 as AUX lane has low power transmitter (LPTX) and low power receiver (LPRX) integrated. And TX0 maps as MIPI datalane1, TX1 maps as MIPI datalane2, TX2 maps to MIPI Clock lane and TX3 maps to MIPI datalane3. Fig1 shows the mapping of TX by eDP or MIPI DPHY configuration. Based on the configuration either eDP controller is enabled or MIPI DSI controller is enabled. The data flow is through the same TX pads but the datarate, swing slope and other electrical parameters are configured based on eDP or MIPI mode.

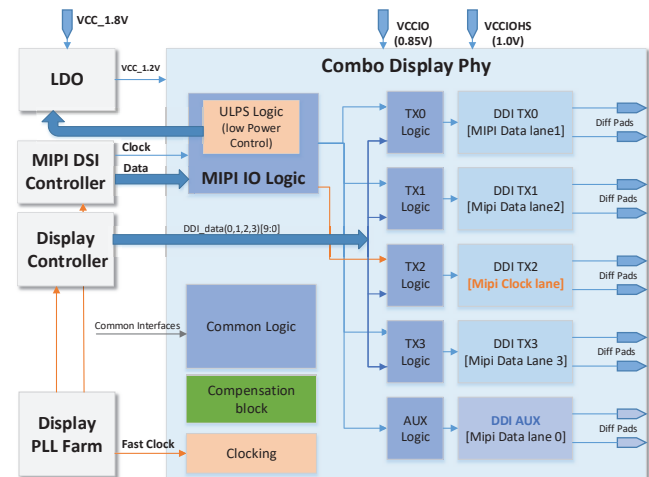


Fig1. Block Diagram of eDP MIPI DPHY PHY Combo Architecture

One primary specification difference is MIPI LPTX and LPRX needs 1.2V signaling. This architecture also integrated LPTX into every datalane and operates from 0v to 1.2V single ended signaling during MIPI low power modes and switched to 1V based high speed mode (HSTX) during data transfer. eDP always operated with 1V high speed transmitters only. In most sub nanometer technology thick gate devices capable of handling 1.2V electrical overstress (EOS) is not available. Further along with the architecture, this paper describes circuit topology to implement combo PHY using thin gate transistors in next section. The circuit Architecture explained is with

thingate stacked transistor transmitter Design for (HSTX) that can support 8.1Gbps data rate.

This feature provides OEMs the flexibility and enable design on the process technologies where Thick Gate devices are not available. LDO based solution enables MIPI latest standard that requires PHY to support LPTX with two different VOH specs based on the Data rate.

III. CIRCUIT ARCHITECTURE

High level Combo TX architecture with stacked thingate solution and the Novel features of our proposal that are implemented in Highlighted yellow blocks shown in Fig 2 and Table I

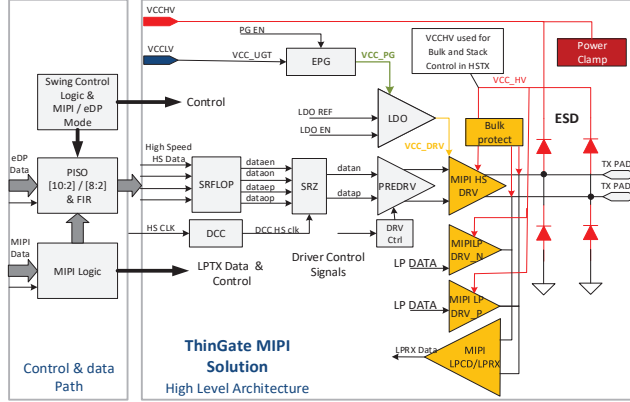


Fig2. Block Diagram of eDP MIPI datablock Architecture

TABLE I: NOVEL FEATURES ON EDP-MIPI COMBO DESIGN

PHY Features	Blocks Implemented	Description
Enabling High performance TX.	HSTX	Stacked differential TX with optimal padcap
Meeting Wide Range Swing Requirement	HSTX + DRIVER LDO	eDP 8.1Gbps needs 600mV swing and MIPI needs 400mV swing.
Enabling High voltage Low Speed TX	LPTX	Stacked single ended TX
Enabling Slew rate Spec	HSTX, LPTX	Bias based slew rate control in pre-driver for HSTX and final LPTX AFE
Enabling EOS Compliant Combophy	HSTX, LPTX, LPRX, HVP	Stacked architecture for TX AFE and high voltage pad tracking circuit for pad leakage reduction

Combophy primarily consists of common clock & data path and common configurable high-speed transmitter (HSTX x4). High Speed transmitter supports data rates up to 8.1 Gbps for eDP and 4.5Gbps as per latest MIPI2.1 specification. For supporting MIPI LPTX, each pad is integrated with stacked LP driver in the same analog IO block. LPTX is disabled in case of eDP operation. The HSTX swings, common mode and AC common mode ripple can be controlled by cleaner programmable supply through inbuilt driver Low Drop Out (LDO).

A. HSTX Architecture with Thin gate

High speed transmitter is shown in fig 3 below. The implementation is a voltage mode differential transmitter with calibrated TX termination. The switches are stacked thingate transistor passgates, designed to operate at swings as low as 400mV differential with a common mode of 200mV. Swing is primarily controlled by driver LDO shown as vccdrv. When HSTX is functioning, the stacked devices connected to pad are turned on completely and the TX switching is controlled by devices in series (the devices with gate named ppdrv and npdrv). LPTX is kept in high impedance state during HSTX operation.

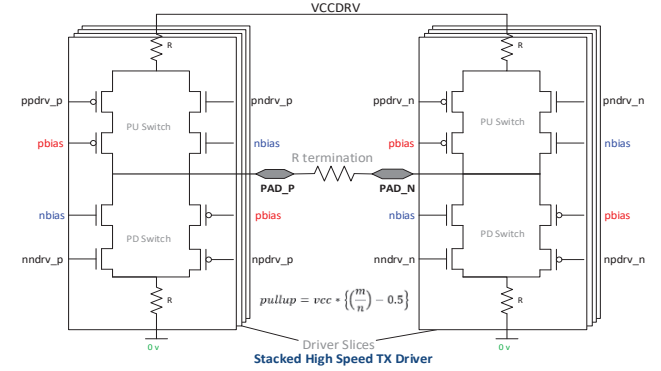


Fig 3. MIPI CKT architecture with thingate and Stacked Driver structure for EOS Solution

When LPTX is operating and the pad toggles at HV voltage, the HSTX is maintained in high impedance state with EOS protection enabled. The Stacked devices of HSTX for PMOS and NMOS are biased appropriately as shown in Fig 3. To maintain the Vgd, Vgs, Vdb and Vsb below the process EOS limits. The sizing of the passgate devices are precisely calculated to finally meet the TX impedance.

B. MIPI Low Power transmitter (LPTX)

MIPI LPTX is used for handshake operations between MIPI panel and MIPI PHY and the signaling is at 10MBPS with VOH requirement by spec of 1.1V to 1.3V. Fig 4 below shows the implementation of a stacked frontend LPTX driver for CMOS signaling powered by VCCHV which is maintained $\geq 1.1V$ and $\leq 1.3V$. The stacked devices in final driver are biased for two purpose. In case of minimum slew rate requirement, the biases are to turn on with maximum overdrive and with gate voltages not to violate EOS limit of the devices. And in case of slew rate control enabled, the stacked devices are throttled to control the drive slopes through pbias_drv and nbias_drv.

The pre-drivers are stacked level shifters that operate on same supply as final driver, supply range limited to $\geq 1.1V$ and $\leq 1.3V$. The primary function of level shifter is to pass the LPTX data from low voltage domain to LPTX driver voltage and serves the pre-driver function to drive LPTX driver switches. The pre-driver transistors are maintained with in EOS limits with constant bias voltage on stacked devices (pbias and nbias shown in fig 4 below). The bias voltages are derived out of a calibrated current reference to keep the bias variation controlled.

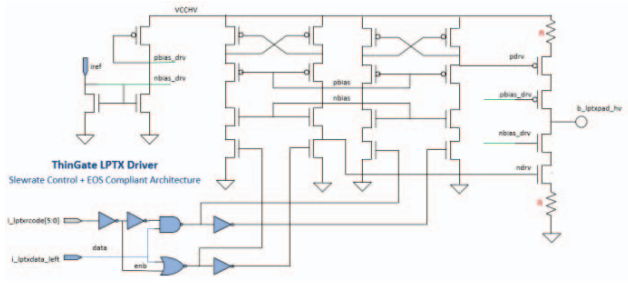


Fig 4. MIPI LPTX architecture with thingate driver, EOS compliant and Slew rate Control

C. Low power RX and Contention Detection(LPRX/LPCD)

MIPI Low Power Receiver (LPRX) is primarily needed to receive sideband communication from far-end LPTX and contention detection is used to detect line contention if both Source and Sink LPTX are transmitting. Conventional Schmitt trigger circuits will not be able to meet the stringent input sensitivity requirements of the low power Contention Detect (LPCD) across different process, voltage, and temperature ranges. A reference voltage based self-biased input stages are used as LPRX and LPCD which are more robust to process, voltage and temperature (PVT) variations. A circuit Architecture for LPRX is as shown in fig 5 below. The VREFHI and VREFLO are adjusted based on the required input thresholds for LPRX and LPCD. The V_{ih} , V_{il} and V_{hys} are dependent on these voltages which are derived from supply voltage using a resistor ladder, hence this scheme would be more robust compared to conventional Schmitt trigger for PVT sensitivity.

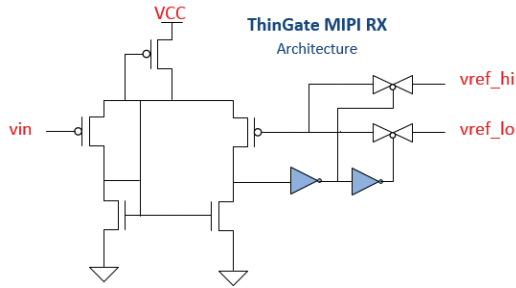


Fig 5. MIPI RX/ LPCD Circuit architecture with EOS protection

D. Bulk protection and biasing circuit

During a scenario where the MIPI lane is in receive mode and the panel side is transmitting the data, the pad voltage can go as high as 1.3V. Since the supply voltage range is lower than this level, there will be bulk violation if the bulk is connected to the supply and pad voltage is higher than bulk. In order to avoid this scenario, a novel circuit is implemented as shown in the fig 6 below. If $V_{PAD} > V_{CC}$, Q1 is ON and $V_{BULK} = V_{PAD}$, if $V_{CC} > V_{PAD}$, Q2 is ON and $V_{BULK} = V_{CC}$. All the bulk nodes of the transistors are connected to the derived voltage V_{BULK} which gets the highest voltage among the V_{PAD} and V_{CC} . This will solve the bulk getting forward biasing when the PAD voltage goes to 1.3V.

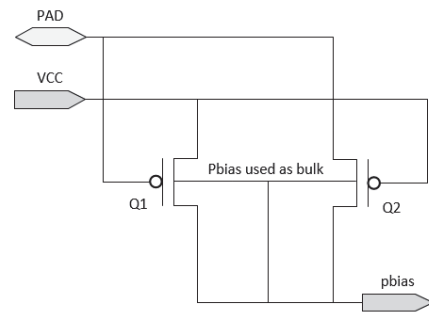


Fig 6. Bulk protection circuit

E. MIPI HSTX Slew rate control circuitry to meet the slew rate specifications

MIPI HSTX slew rate control circuitry plays a vital role to meet the slew rate specifications MIPI DSI to retain the EMI/EMC under control. Maintaining slower slopes (150ps to 233ps) for MIPI data rates below 1Gbps and faster slopes for eDP 5.4Gbps (<50ps) & higher data rates was a challenge with common TX. Fig 7 shows the innovative solution that controls the slope of the pre-driver, by controlling the stacked device currents. A programmable bias generates control voltage during MIPI DSI mode and turns on the stack transistor with full overdrive during eDP mode. The throttled currents on pre-driver maintains slower slopes on pre-driver output and hence controlling final driver slopes

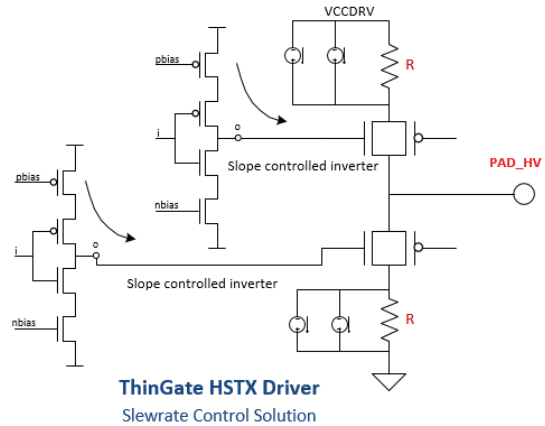


Fig 7. Slew rate control concept for HSTX

To meet swing and common specifications DP and MIPI high speed voltage mode driver is powered by LDO which outputs 0.4V. In order to support high speed, thin gate switches are used. This 0.4V power is used to majorly to meet the MIPI's VCM of 200 mV and swing requirement of 200 mV. As voltage is lower, instead of regular pmos pull up, nmos pull up switches are used while pull down switches are regular nmos devices. Data is preconditioned in the pre driver which is powered by power gate outputting 1.05V supply which helps in providing sufficient overdrive to turn on and off the driver switches.

IV. POST SILICON RESULTS

The architecture and circuit topology described in this paper not only integrates MIPI DPHY & eDP on single port but also provides solution to MIPI high voltage signaling and configurable slew rate control for HSTX and LPTX. Below fig7 and fig8 are post silicon results for the eDP at 8.1Gbps and MIPI operating at 4.5Gbps respectively. The results are circuit architecture that was implemented in 10nm FinFet technology. The design meets eDP compliance and MIPI conformance.

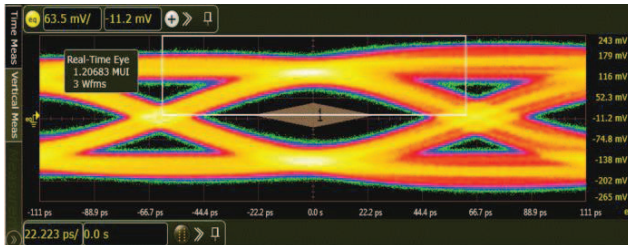


Figure7: eDP 8.1Gbps, 80C, 0.95V, 11dB channel

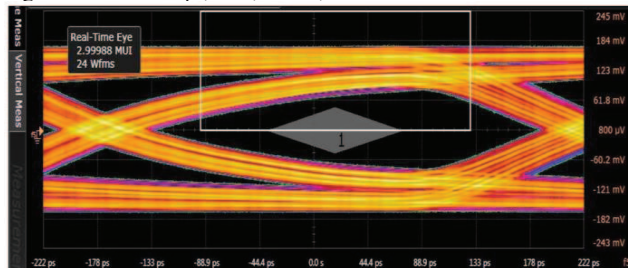


Figure8: MIPI DSI 4.5Gbps, 25C, 0.85V, 6 dB channel

V. CONCLUSION

eDP-MIPI Combophy with thin gate circuit Architecture is proven to support eDP 8.1Gbps (eDP1.5 specification) and MIPI at 4.5Gbps (DPHY2.0). The result shows the implementation works up to max specification including features like slewrate control, EOS compliance and PHY configurability. The implementation provides flexibility to the OEMs to configure using the same port on SOC either to eDP or MIPI DSI panels even at the last minute. Usage of eDP-MIPI combophy will result in significant area savings for (~27% in 10nm process). Reduction in number of bumps/pins (10 Number), which will result in package and silicon cost saving. This implementation helps SOC to make smaller form factor designs. Usage of combophy helps in reducing design time significantly and hence the "Time to Market" which is very important in the SOC market trends.

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